LocalCloudManager

System Description

Abstract

This document describes the LocalCloudManager support system of the Eclipse Arrowhead 5.2.2 framework, as implemented in `ah5-LocalCloudManager-java-spring` of the primary Java source tree (`Eclipse-Arrowhead-java`) and deployed with the standalone Compose definition of that folder.

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1 Overview

This document describes the LocalCloudManager system. LocalCloudManager is the operator console of the local cloud: it shows the state of the cloud and carries out the management actions an operator starts, with its own Arrowhead identity.

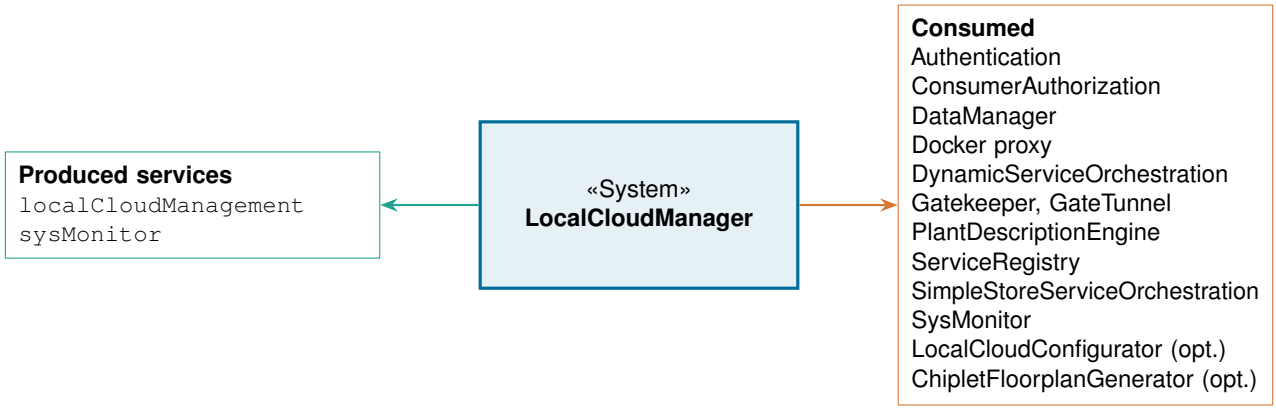


Figure 1: SysML-style block view of LocalCloudManager with its produced and consumed services.

The rest of this document is organized as follows. In Section 1.1, we reference major prior art capabilities of the system. In Section 1.2, we describe the intended usage of the system. In Section 1.3, we describe fundamental properties provided by the system. In Section 1.4, we describe delimitations of capabilities of the system. In Section 2, we describe the abstract service functions consumed or produced by the system. In Section 3, we describe the security capabilities of the system.

1.1 Significant Prior Art

New in this workspace.

1.2 How This System Is Meant to Be Used

The operator opens the console (`http://127.0.0.1:8490`), logs in with the console password and works in the views (overview, registry, orchestration rules and authorization policies, plant descriptions, SoS composition, consumers, data, sysMonitor, inter-cloud, runtime and, when enabled, local-cloud configuration and chiplet floorplans).

1.3 System functionalities and properties

1.3.1 Functional properties of the system

- Views assembled from the core and support systems.
- Orchestration rules (create, remove, change priority), plant descriptions and sysMonitor subscriptions.
- Local-cloud management authorization policies of ConsumerAuthorization (view, grant, replace, revoke), with optimistic concurrency on the expected policy snapshot.
- System-of-Systems (SoS) compositions, stored as plant descriptions in the PlantDescriptionEngine; saving an active SoS needs an explicit confirmation because it changes orchestration.
- Optional graphical local-cloud configuration through the LocalCloudConfigurator (catalog, saved designs, plans and reviewed deployments) when `LOCAL_CLOUD_CONFIGURATOR_URL` is set.
- Optional chiplet floorplan editing through the ChipletFloorplanGenerator (orchestrated) when `CHIPLET_FLOORPLANS_ENABLED=true`.
- Container logs and lifecycle through the Docker proxy.
- Native sysMonitor observations.

1.3.2 Configuration of system properties

The system is configured with environment variables. The variables that shape its behaviour are:

Property	Default	Meaning
`PORT`	8491 (API), 8490 (UI)	Ports.
`SERVICE_REGISTRY_URL`	`https://serviceregistry:8443`	Registry.
`TLS_CA_CERT` / `TLS_CLIENT_*`	(required)	Cloud CA and LocalCloudManager credential.
`CONSOLE_PASSWORD_HASH_FILE` / `CONSOLE_ALLOWED_HOSTS`	(required)	Console login and accepted host names.
`MANAGED_SERVICES`	(list)	Containers the console may start, stop and restart.
`DOCKER_API_URL` / `COMPOSE_PROJECT_NAME`	(required)	Docker proxy (standalone deployment: no proxy, `MANAGED_SERVICES` empty).
`CONSOLE_PROXY_SECRET_FILE`	(required)	Secret shared with the console's nginx; API requests without it are refused (403).

Property	Default	Meaning
`CONSOLE_SESSION_SECONDS`	28800	Console session lifetime.
`AUTHENTICATION_PASSWORD_FILE`	(required)	Identity password of LocalCloudManager.
`TLS_CRL_FILE`	(unset)	CRL of the cloud CA for provider certificates, fail closed (standalone deployment: `/crl/ca.crl`).
`ARROWHEAD_SYSTEM_NAME` / `ORCHESTRATOR_SYSTEM`	LocalCloudManager / DynamicServiceOrchestration	Own system name; orchestrator looked up in the ServiceRegistry.
`ORCHESTRATION_TTL_SECONDS`	300	Reuse of orchestration results.
`ORCHESTRATION_PRIORITY_MODE`	(native)	`replace`: change a rule priority by create and remove instead of SimpleStore `modify-priorities`.
`SYSMONITOR_BROKER_*` / `SYSMONITOR_LOCAL_CLOUD`	amqp-broker / 5671	Native sysMonitor publisher.
`LOCAL_CLOUD_CONFIGURATOR_URL` / `_TOKEN_FILE`	(unset)	Enables the local-cloud configuration view; bearer token of the configurator.
`CHIPLET_FLOORPLANS_ENABLED`	false	Enables the chiplet floorplan view (console name ChipletLocalCloudManager).

Secrets (identity password, console password hash, console proxy secret, configurator token, private key) have no usable defaults; the standalone deployment reads them from per-deployment Compose file secrets (`*_FILE` variables) under `/run/secrets`.

1.3.3 Data stored by the system

None durable; sessions and the identity token in memory. SoS compositions are stored by the PlantDescription Engine, policies by ConsumerAuthorization and designs by the LocalCloudConfigurator.

1.3.4 Non functional properties

A failing provider gives a partial view with an error note.

1.3.5 Stateful or stateless

Stateless apart from sessions.

1.4 Important Delimitations

Acts only within the grants given to LocalCloudManager (not a sysop since J23).

2 Services

This section describes the produced and consumed services. Each produced service is specified by an SD document and one IDD document per interface.

2.1 Produced services

Service	Version	Interfaces	SD	IDD
`localCloudManagement`	1.0.0	HTTP(S)	[1]	[2]
`sysMonitor`	1.0.0	AMQP(S)	[3]	[4]

The console API is registered with `generic_http` and policy `NONE`; `GET /health` and the console endpoints added after the registration list (features, authorization policies, rule priority, SoS, local-cloud configuration, chiplet floorplans) are not registered; the IDD specifies them.

2.2 Consumed services

Provider	Service	Operations	Use	Condition
ServiceRegistry	serviceDiscovery; serviceRegistry Management	register, lookup; queries	Registration; inventory.	always
Authentication	identity	identity-login	Own token.	always
DynamicService Orchestration	service Orchestration; service Orchestration History Management	pull; query	Orchestration of all other services; history.	always
SimpleStore Service Orchestration	service Orchestration SimpleStore Management	query, create, remove, modify-priorities	Rules and rule priorities.	grant
Consumer Authorization	authorization Management	query-policies, grant-policies, revoke-policies	Consumers view; policy editor.	grant
DataManager	dataHistorian	list-systems, get-data	Data view.	grant
SysMonitor	sysMonitorHistory; sysMonitor Management	all	sysMonitor views.	grant
PlantDescription Engine	plantDescription Management	all	PD and SoS views.	PDE whitelist with LocalCloudManager
Gatekeeper, GateTunnel	gatekeeper Management; gate TunnelManagement	status, sessions	Inter-cloud view.	grant
Docker proxy	Docker Engine subset	list, logs, start/stop/restart	Runtime view.	proxy deployed

Provider	Service	Operations	Use	Condition
LocalCloud Configurator	configurator REST API (bearer token, not orchestrated)	catalog, status, configurations, plan, deployments	Local-cloud configuration view.	configurator URL set
ChipletFloorplan Generator	floorplanDiscovery; floorplan Management	query-floorplans, get-floorplan; create, replace, remove, check-network	Floorplan view.	floorplans enabled, grant

Only Authentication's `identity` and the orchestrator are looked up in the ServiceRegistry; other Arrowhead services are reached through DynamicServiceOrchestration (results cached), and every such call needs a ConsumerAuthorization grant.

3 Security

LocalCloudManager is a Node.js console, not a Spring system. Towards the cloud it always uses HTTPS with its X.509 certificate following the Arrowhead certificate profile (`LocalCloudManager.<Cloud>.<Organization>.arrowhead.eu`); provider certificates not issued under the cloud CA, or revoked in its CRL, are refused. The operator side is plain HTTP on the host's loopback address through the console's nginx. The standalone deployment of this folder (`compose.yaml`) runs this configuration.

3.1 Security model

- **Protocols:** HTTPS (TLS 1.3 and 1.2) with its client certificate towards the cloud; AMQP over TLS for `sysMonitor`; plain HTTP between the browser, nginx and the API on the loopback address and the console's own networks.
- **Data protection:** transport encryption by TLS towards the cloud; no database. Secrets are kept out of the container environment (Compose file secrets).
- **Operator authentication:** console password (scrypt hash), HttpOnly SameSite=Strict session cookie, lock-out after repeated failures, allowed host names.
- **Towards the cloud:** mutual TLS with the LocalCloudManager certificate and an identity token bound to it; rights are exactly the grants given to LocalCloudManager.
- Console access: only requests forwarded by the console's nginx (proxy secret), allowed host names (against DNS rebinding), scrypt-hashed password, JSON-only login, HttpOnly SameSite=Strict session cookie, per-client lock-out after five wrong passwords, `X-Arrowhead-Action: confirm` on changes and expected-snapshot checks (409) for policy, rule-priority, SoS and floorplan edits. Docker only through the proxy.

For the Arrowhead certificate profile see github.com/eclipse-arrowhead/documentation.

4 References

[1] J. Delsing, "localCloudManagement – Service Description (LocalCloudManager)," Eclipse Arrowhead, SD 5.2.2, 2026, localCloudManager-v5.2.2-localCloudManagement-SD.pdf.

[2] ——, "localCloudManagement HTTP/TLS/JSON – Interface Design Description (LocalCloudManager)," Eclipse Arrowhead, IDD 5.2.2, 2026, localCloudManager-v5.2.2-localCloudManagement-HTTP-IDD.pdf.

[3] ——, "sysMonitor – Service Description (LocalCloudManager)," Eclipse Arrowhead, SD 5.2.2, 2026, localCloudManager-v5.2.2-sysMonitor-SD.pdf.

[4] ——, "sysMonitor AMQP/TLS/JSON – Interface Design Description (LocalCloudManager)," Eclipse Arrowhead, IDD 5.2.2, 2026, localCloudManager-v5.2.2-sysMonitor-AMQP-IDD.pdf.

5 Revision History

5.1 Amendments

No.	Date	Version	Subject of Amendments	Author
1	2026-10-22	5.2.2	Document created from the Arrowhead LaTeX templates for the 5.2.2 Java Spring implementation, including the local extensions of this workspace.	Jerker Delsing
2	2026-10-30	5.2.2	Checked against the primary Java source tree (Eclipse-Arrowhead-java): console functions, configuration, consumed services and security updated to the current Node.js implementation (policies, rule priority, SoS, local-cloud configuration, chiplet floorplans).	Claude Opus 5.5 (draft)

5.2 Quality Assurance

No.	Date	Version	Approved by
1		5.2.2	